

Exploring isolated neutron stars from pulsars to magnetars with NewAthena[★]

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Abstract

Isolated neutron stars (INSs) comprise several classes, including rotation-powered pulsars (RPPs), X-ray Dim Isolated Neutron Stars (XDINSs), magnetars and Central Compact Objects (CCOs). They emit X-rays with spectra typically described by a thermal component tracing neutron star cooling and a non-thermal power-law tail from magnetospheric processes, offering a unique probe of ultra-dense matter and strong magnetic fields. NewAthena's unprecedented spectral resolution and large collecting area will transform this field by significantly expanding the sample of X-ray-detected INSs and enabling precise measurements of luminosity variability and secular spin-down across classes, directly testing magneto-thermal evolution models. Detailed characterisation of complex X-ray pulsations will constrain non-dipolar surface magnetic fields and their role in powering emission, extending such studies to CCOs despite their low pulsed fractions. High-quality spectroscopy will disentangle surface emission mechanisms (e.g., magnetised atmospheres versus condensed surfaces), measure hotspot sizes and locations, and characterise deviations from the thermal continuum, such as broad absorption features and cyclotron lines. The mission will also test emerging evidence for faint non-thermal components in XDINSs. For magnetars, high-cadence phase-averaged and phase-resolved spectroscopy during outbursts will trace long-term pulse-profile and spectral evolution, revealing hotspot formation and outburst decay, while the detection of proton and electron cyclotron features will directly measure local magnetic fields, probing magnetic topology beyond spin-down constraints and advancing a unified picture of neutron star physics.

Keywords: NewAthena, Neutron stars, Rotation-powered pulsars, Central compact objects, X-ray dim isolated neutron stars, magnetars

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1. Introduction

Since their discovery, isolated neutron stars (NSs) have exhibited a remarkable variety of behaviours depending on their age, magnetic field strength and geometry, rotational dynamics, emission mechanisms, and surrounding environments. This diversity in their manifestations has led astronomers to classify isolated NSs (INSs) into numerous categories, as shown in the spin period P – spin period derivative $\dot{P}$ diagram in Fig. 1 (for a review, see e.g., Borghese and Esposito, 2023, and references therein). As of September 2026, the ATNF Pulsar Catalogue v2.8.1 lists 4393 pulsars, of which $\sim 80\%$ are expected to be non-recycled (i.e. not spun up by accretion in a binary system, $P > 20$ ms and $\dot{P} > 10^{-18}$ s s $^{-1}$; Manchester et al., 2005). Hereafter, we will focus on non-recycled INSs.

Rotation-powered pulsars (RPPs) account for the vast majority of known INSs, in part because their coherent radio emission makes them comparatively easy to identify in traditional pulsar surveys. Pulsed emission from RPPs is detected across the electromagnetic spectrum, from radio to X-ray and gamma-ray energies (Manchester et al., 2005; Smith et al., 2023). Their X-ray efficiencies, defined as the ratio of X-ray luminosity to spin-down power, $\eta_X = L_X/\dot{E}$, typically span $\sim 10^{-5}$ – 10^{-3} . The pulsar X-ray emission generally comprises a non-thermal magnetospheric component and thermal emission from the NS surface, while in the most energetic systems an unresolved or surrounding pulsar wind nebula can provide an additional, and sometimes dominant, non-thermal contribution. Sensitive phase-resolved spectroscopy can constrain the emission region of the non-thermal component and its connection to the gamma-ray-emitting region. Thermal emission can arise from both the cooling NS surface and hotter regions heated by crustal magnetic fields and magnetospheric return currents. It becomes increasingly prominent in middle-aged and older pulsars as the magnetospheric contribution fades. Phase-resolved spectroscopy combined with realistic atmosphere models can help disentangle these thermal components and determine how their relative contributions depend on NS age and spin-down power. It can also clarify the origin of phase-dependent spectral residuals reported in several RPPs, including PSR B0656+14 (Aru-mugasamy et al., 2018), PSR J1740+1000 (Kargaltsev et al., 2012a), and PSR J1412+7922 (Mereghetti et al., 2021), although such features are not ubiquitous across the population.

X-ray dim INSs (XDINSs) are a local (< 0.5 – 1 kpc) population of NSs, characterised by the absence of associated supernova remnants (SNRs), pulsar wind nebulae, and coherent pulsed radio or gamma-ray emission. XDINSs are observed to emit primarily in soft X-rays, with blackbody-like thermal spectra characterised by temperatures of $kT \sim 45$ – 110 eV and low hydrogen column densities ($N_H \sim 10^{20}$ cm $^{-2}$). Over the past decades, broad and narrow absorption features have been discovered in the phase-averaged and phase-resolved X-ray spectra (e.g. Hohle et al., 2012; Borghese et al., 2015). The physical origin of these features remains uncertain: they might be due to proton cyclotron resonances, atomic transitions in a magnetised atmosphere or arise from an inhomogeneous surface temperature distribution (Zane et al., 2001; Lai, 2001; Viganò et al.,

2014). The X-ray emission is pulsed at spin periods of a few seconds ($P \sim 3$ – 11 s), with spin-down rates of $\dot{P} \sim 10^{-14}$ s s $^{-1}$, implying surface dipolar magnetic fields of $B_{\text{dip}} \sim 10^{13}$ G (see Bogdanov and Ho, 2024, and references therein).

Magnetars are relatively slow rotators ($P \sim 1$ – 12 s) and exhibit large spin-down rates ($\dot{P} \sim 10^{-13}$ – 10^{-11} s s $^{-1}$). These timing properties translate into dipolar fields $B_{\text{dip}} \sim 10^{14}$ – 10^{15} G, making magnetars the strongest magnets in the Universe. The current magnetar census comprises about 30 sources. Their persistent X-ray luminosity, $L_X \sim 10^{32}$ – 10^{36} erg s $^{-1}$, is generally larger than their spin-down luminosity, making the decay and instabilities of the super-strong magnetic field the main engine of their emission. The persistent soft X-ray (< 10 keV) spectra are modelled with a thermal blackbody component ($kT \sim 0.3$ – 1 keV) and, in some cases, an additional component is required, typically a second blackbody ($kT \sim 1$ – 2 keV) or a power law (photon index $\Gamma \sim 1$ – 2). The distinctive trait of magnetars is their transient activity over a wide range of timescales and luminosities, from short bursts lasting milliseconds to seconds and reaching $L_X \sim 10^{36}$ – 10^{43} erg s $^{-1}$, to rare giant flares, whose sub-second initial spikes can reach peak luminosities of $\sim 10^{44}$ – 10^{47} erg s $^{-1}$ and are followed by pulsating tails lasting hundreds of seconds. Magnetars also undergo outbursts, during which the persistent X-ray luminosity can increase by up to three orders of magnitude and subsequently decay back in most cases towards the quiescent level over months to years. During an outburst, the X-ray spectrum initially hardens and then gradually softens on the timescales of the luminosity relaxation. Short bursts might also have radio counterparts: SGR J1935+2154 emitted a fast radio burst (FRB)-like radio burst coincident with an X-ray burst (CHIME/FRB Collaboration, 2020).

Central compact objects (CCOs) are soft X-ray emitters (~ 0.2 – 5 keV) located near the centres of young SNRs. They lack surrounding pulsar wind nebulae and, with one notable exception, have no detected counterparts at other wavelengths (De Luca, 2017; Zhang et al., 2026). While only a dozen CCOs are known, their locations in young SNRs suggest they may represent a significant fraction of the NS population. Their thermal spectrum is well described by the sum of two blackbodies with temperatures $kT \sim 0.2$ – 0.5 keV and small emitting radii spanning from 0.1 to a few kilometres. Their emission is usually steady with a luminosity of the order of 10^{33} erg s $^{-1}$; pulsations are detected in three CCOs at periods in the range ~ 0.1 – 0.4 s. For these three objects, spin-down measurements imply dipole magnetic fields of only $\sim 10^{10}$ – 10^{11} G, among the weakest measured for young INSs, at odds with their relatively bright thermal emission. One CCO instead showed magnetar-like outbursts and a ~ 6 hr modulation interpreted as a long spin period (De Luca, 2017; Rea et al., 2016).

This overall picture has become more complex with the discoveries of magnetar-like outbursts from two high- B RPPs (e.g. Gavriil et al., 2008) and one CCO, while three magnetars have dipolar magnetic fields within the range of those of ordinary RPPs (e.g. Rea et al., 2010), and radio emission has been detected from a few magnetars. These results suggested evolutionary links among the different INS classes, a possibility reinforced by the fact that the sum of the individual birthrates of the

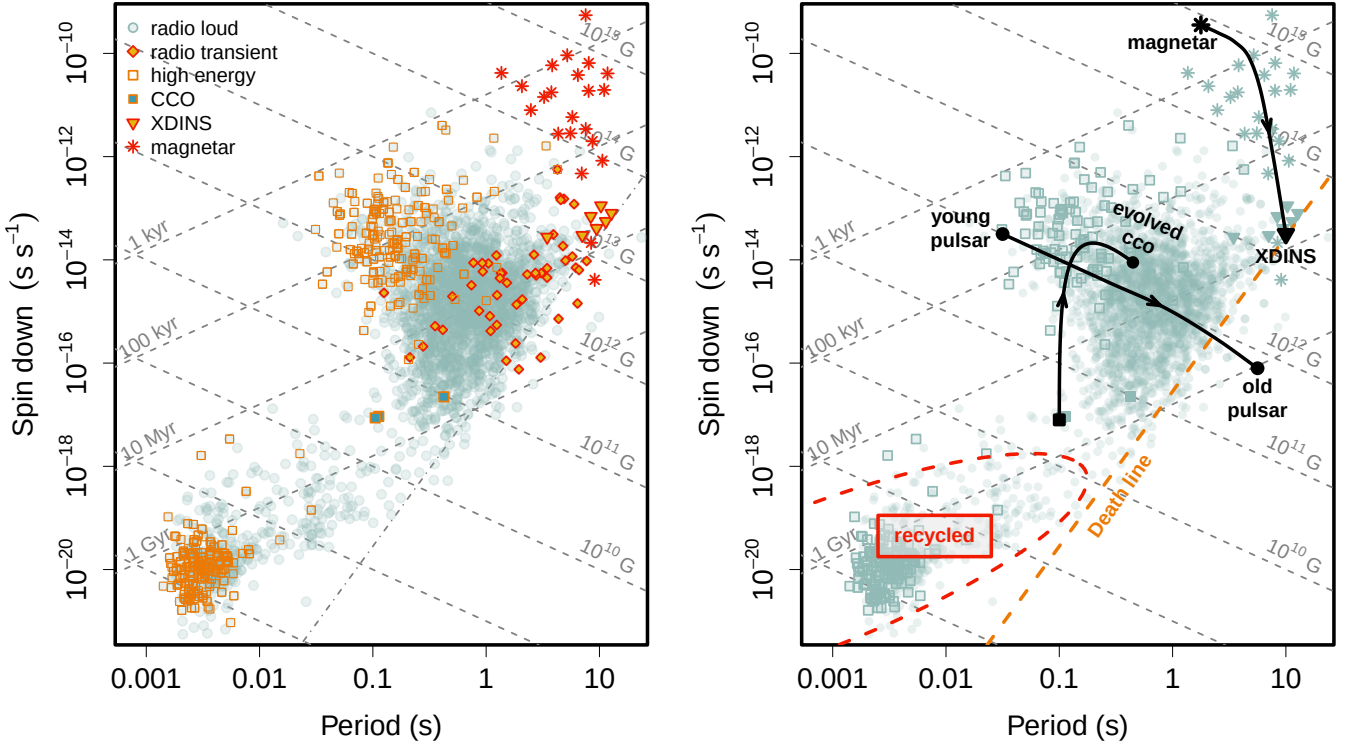


Figure 1: $P - \dot{P}$ diagram of the Galactic NS population. Dashed lines show lines of constant characteristic age ($\tau_c \sim P/2\dot{P}$) and constant dipolar field ($B_{\text{dip}} \approx 3.2 \times 10^{19} (P\dot{P})^{1/2} \text{ G}$). These two quantities are estimated under standard vacuum-dipole assumptions (i.e., orthogonal rotator in vacuum with $R = 10 \text{ km}$ and $I = 10^{45} \text{ g cm}^2$). The position of peculiar sources is highlighted against the background of high-energy and radio spin-powered pulsars (see legend). Right: Schematic view of pulsar evolution scenarios in the diagram. While all NSs experience magnetic field decay, its timescale depends strongly on the initial field: magnetars decay within tens of kyr, evolving towards XDINSs, whereas classical RPPs evolve close to lines of constant inferred field for their first hundreds of kyr before their fields also decay (e.g. Viganò et al., 2013). CCOs may instead undergo field growth through the re-emergence of a buried field (e.g. Ho, 2015).

INS populations exceeds the Galactic core-collapse supernova rate (Keane and Kramer, 2008).

Further evidence for such connections comes from 2XMM J104608.7–594306, a thermal source with properties similar to those of XDINSs, but without a firmly established spin period (Pires et al., 2015). The recent detection of unusual bursts of coherent radio emission (e.g. Rajwade et al., 2025) suggests a possible link to radio-loud magnetars or repeating FRBs (CHIME/FRB Collaboration, 2020)). Similarly, the discovery of faint radio pulsations from a CCO (Zhang et al., 2026) further blurs the boundaries between the various classes of INSs. Theoretical studies have been put forward, highlighting the magnetic field and its evolution as key ingredients to explain the puzzling diversity and the evolutionary paths of the INS population (see e.g., Viganò et al., 2013; Dehman et al., 2023; Ascenzi et al., 2024; Gourgouliatos et al., 2016). For instance, according to magneto-thermal evolutionary models, XDINSs may be descendants of magnetars where the magnetic field had already decayed. However, further observational constraints are required, and NewAthena will be uniquely suited to provide them through its combination of high sensitivity, angular resolution, large effective area, and low background.

2. Rotation-powered pulsars

RPPs act as unipolar inductors, extracting rotational energy to generate large-scale electric fields that accelerate particles and power magnetospheric radiation. Their X-ray emission can be broadly divided into two physically distinct but potentially connected components: non-thermal emission from relativistic particles in the magnetosphere, and thermal emission from the NS surface. The latter can be powered predominantly by long-term cooling in young RPPs, while it can be increasingly dominated by return-current heating and crustal field decay in the polar caps in older RPPs.

In the standard pair-cascade framework (Ruderman and Sutherland, 1975), particles accelerated above the polar cap to Lorentz factors of $\gamma \sim 10^6 - 10^7$ emit high-energy curvature (and, in some regimes, inverse-Compton) photons that initiate magnetic pair production. The resulting electron-positron cascade supplies the magnetospheric plasma, while returning particles heat the NS surface and produce localized thermal X-ray emission. Thus, non-thermal and thermal X-ray components are linked through the pair-cascade process: the non-thermal emission constrains particle acceleration and magnetospheric plasma physics, while the thermal emission probes the surface magnetic structure and return-current heating.

Despite decades of study, one of the fundamental assumptions underlying pulsar pair-cascade theory remains poorly con-

strained: the existence of strong, small-scale multipolar magnetic fields anchored in the NS crust. Although spin-down measurements only probe the dipolar component (the multipoles are negligible at the light cylinder), the magnetic topology within a few stellar radii may contain significantly stronger and more complex structures. Such non-dipolar surface magnetic fields have been invoked based on several observational and theoretical considerations:

1. **Polar-cap sizes inferred from thermal X-ray emission:** Thermal X-ray observations of RPPs often imply emitting areas substantially smaller than the canonical polar-cap area expected for a dipolar magnetic field extending down to the NS surface (Rigoselli and Mereghetti, 2018). If the X-ray emitting region is associated with the magnetic polar cap and magnetic flux conservation is assumed, a smaller surface area requires a stronger local magnetic field than the large-scale dipole component inferred from pulsar spin-down. This provides indirect evidence for enhanced, non-dipolar magnetic fields near the stellar surface.
2. **Phase offsets between radio and X-ray emission:** The observed phase offsets between radio pulses and thermal X-ray emission from polar caps are often inconsistent with expectations from a purely dipolar magnetic geometry (e.g. Arumugasamy and Mitra, 2019; Pétri and Mitra, 2020). Because the radio and thermal X-ray components are expected to be tied to the open magnetic-field region, the measured offsets provide a direct constraint on the near-surface magnetic topology. Larger-than-expected offsets indicate that the magnetic pole locations and field-line geometry close to the NS surface may be significantly distorted by strong multipolar magnetic components.
3. **Absorption features in the thermal X-ray spectrum:** Absorption features detected at ≈ 100 eV may originate from proton cyclotron resonances on a magnetic field of strength $B \approx 1.6 \times 10^{13} (1 + z) \left(\frac{E_{\text{cyc}}}{100 \text{ eV}} \right) \text{ G}$ (where z is the gravitational redshift at the line-forming region) or by atomic transitions at similar field strengths. Even though this condition is usually met on the surface of magnetars and XDINSSs, such features have been recently detected also in the X-ray spectra of RPPs, which have a weaker dipolar field of the order of 10^{12} G (e.g. Arumugasamy et al., 2018). As suggested in Section 4.1.2, this may be due to local enhancements of the B -field, possibly caused by small-scale multipolar structures near the stellar surface.
4. **Requirements from radio emission and pair cascades:** Coherent radio emission from pulsars requires sufficiently large pair multiplicities in the magnetosphere. Modern pair-cascade simulations (Timokhin, 2010) indicate that a purely dipolar magnetic geometry often struggles to produce the required particle supply, especially in older RPPs approaching the radio death line (Geppert et al., 2013; Mitra et al., 2020). Strong small-scale multipolar fields near the surface can enhance the curvature of magnetic field lines, reducing their curvature radius and increasing

the efficiency of curvature-radiation photon production and magnetic pair creation. Such localised magnetic structures therefore provide a natural mechanism for sustaining pair cascades when a purely dipolar field is inadequate.

5. **Plasma instability requirements for coherent radio emission:** The generation of coherent radio emission requires plasma instabilities in the relativistic electron-positron plasma flowing along open magnetic field lines. These instabilities are sensitive to the plasma distribution and are suppressed when the plasma becomes hot (Buschauer and Benford, 1977). Strong small-scale multipolar magnetic fields near the stellar surface provide a natural way to regulate these conditions by increasing the relative streaming between the electron and positron components in the pulsar pair plasma (Rahaman et al., 2020).
6. **Polar-cap heating and sub-pulse drifting constraints:** Observations of drifting sub-pulses indicate that the polar-cap accelerator is unlikely to operate as a fully unscreened vacuum gap. Instead, partial screening of the accelerating electric field by thermally emitted ions or other charge carriers is required to reproduce the observed drift rates. Strong near-surface multipolar fields provide such conditions by regulating particle acceleration in the polar-cap region (Gil et al., 2003).

However, inferred polar-cap temperatures and emitting areas are subject to substantial uncertainties. To date, thermal polar-cap emission has been detected only from a small sample of nearby (see Fig. 10 of Rigoselli and Mereghetti 2018), relatively bright RPPs, preventing us from determining whether concentrated polar caps and enhanced surface magnetic fields are generic features of RPPs or occur only in special systems.

NewAthena will significantly advance this field through the large collecting area and wide field of view of WFI, enabling the discovery and characterization of more RPPs through improved sensitivity and serendipitous detections. For relatively bright RPPs, the high spectral resolution of X-IFU will enable detailed searches for absorption features and provide new constraints on their surface emission and atmosphere. These capabilities will enable sensitive measurements of faint thermal polar-cap emission from a much larger population of older RPPs, together with detailed characterisation of their non-thermal magnetospheric spectra, which may also help to examine the reported anti-correlation between fluxes and photon indices of non-thermal X-ray emissions from pulsars (Chang et al., 2023). Phase-resolved spectroscopy combined with magnetised atmosphere modelling will constrain polar-cap luminosities, temperature distributions and location of the emitting areas, as well as detect absorption features. Tightly constrained X-ray spectra and polar-cap sizes can provide quantitative tests of the interplay between the magnetic-field geometry of the NS, return-current heating, and atmosphere physics in RPPs.

However, thermal emission alone cannot uniquely determine the pair-cascade physics. The non-thermal X-ray component may also involve processes such as inverse-Compton scattering of thermal seed photons, making the connection between surface

heating and magnetospheric radiation essential. Correlations between thermal polar-cap properties and the non-thermal spectrum will therefore provide direct constraints on pair-production efficiency and particle acceleration. Mode-changing pulsars such as PSR B0943+10 (Hermsen et al., 2013; Rigoselli et al., 2019b) provide particularly powerful laboratories for this investigation. Their abrupt changes in radio and X-ray emission, spin-down torque, and magnetospheric state are widely interpreted as transitions in the global current system and pair-production activity. If the thermal and non-thermal components are coupled through magnetospheric current closure, changes in the non-thermal emission should be accompanied by variations in polar-cap heating. NewAthena will enable sensitive X-ray studies across different magnetospheric states, directly testing the coupling between surface magnetic topology, pair production, and magnetospheric currents.

NewAthena will therefore provide the first population-scale test of the multipolar magnetic-field hypothesis underlying modern pulsar pair-cascade and radio-emission theories, while also testing whether X-ray and radio mode changes are synchronous in a larger pulsar sample.

3. X-ray dim isolated neutron stars

For many years, the ROSAT-discovered “Magnificent Seven” represented the main sample of XDINSs (e.g. see Turolla, 2009, and references therein). Their predominantly thermal X-ray emission, largely free from magnetospheric contamination, makes them valuable laboratories for studying cooling NS surfaces and testing NS cooling and emission models. Spectral modelling and absorption features detected in many XDINSs also provide constraints on NS radii and compactness, offering insights into the equation of state of cold nuclear matter (e.g. Ascenzi et al., 2024).

Several open questions remain that NewAthena can help address. For instance, it is still unclear whether their radio-quiet nature is an intrinsic property or instead results from an unfavourable viewing geometry (Kondratiev et al., 2009; Rigoselli et al., 2019a). NewAthena will enable more detailed phase-resolved spectral studies and improved pulse-profile modelling, providing tighter constraints on viewing geometries and impact parameters. Similarly, X-IFU’s high spectral resolution will enable detailed studies of absorption features, helping to clarify their still uncertain physical origin.

A further question is whether faint hard X-ray emission is a common property of XDINSs and what produces it. A power-law component emerging above ~ 2 keV has been detected in one XDINS, with photon index $\Gamma \approx 1.4$ and a 2–8 keV flux of $\approx 2.5 \times 10^{-15} \text{ erg cm}^{-2} \text{ s}^{-1}$ (De Grandis et al., 2022a). Sensitive WFI observations can determine whether similar hard tails are present in other XDINSs and constrain their spectral shape, helping to distinguish non-thermal emission from additional hot thermal components. Establishing how frequently such tails occur, and comparing their luminosities with the spin-down power, would test whether residual magnetospheric particle acceleration is a common feature of XDINSs and whether it can account

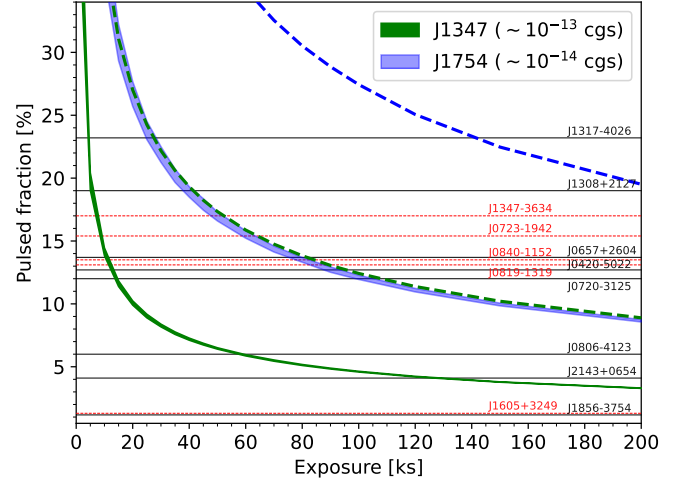


Figure 2: Observable pulsed fraction limits (5σ detection threshold) for NewAthena WFI (solid) and XMM-Newton EPIC-pn (dashed; median of simulations) as a function of exposure time for one recently discovered thermally emitting source, J1347 (eROSITA), and the faintest candidate J1754 (XMM-Newton). Curves are based on XSPEC simulations of 1000 spectra using the corresponding WFI (optical blocking filter) and EPIC-pn (THIN filter) responses. Horizontal lines indicate measured pulsed fractions (black) or upper limits (red dotted) for the “Magnificent Seven” and eROSITA INSs.

for their hard X-ray emission, providing a direct link to the magnetospheric activity seen in RPPs.

The local XDINS birthrate is comparable to that of the much more numerous RPP class, suggesting that XDINSs may represent a significant fraction of the Galactic NS population. Constraining their birthrate is therefore key to addressing the NS birthrate problem, calibrating population synthesis models, and testing models of INS magneto-thermal evolution (Pardo-Araujo et al., 2026; Ronchi et al., 2026). Simulations show that, for the same synthetic population and exposure law, NewAthena will reach in ~ 5 – 7 years a discovery space comparable to that accumulated by XMM-Newton over more than two decades.

Recent searches in the XMM-Newton observational footprint and the newly conducted eROSITA All-Sky Survey (eRASS; e.g. Kurpas et al., 2024; Pires et al., 2026) have led to the first XDINS discovery in more than two decades (Kurpas et al., 2026a) and to the identification of at least 11 new thermally emitting INSs or high-likelihood candidates of unknown NS nature ($d \sim 1 - 7$ kpc; Pires et al., 2026; Kurpas et al., 2026b). NewAthena will be crucial for characterising this growing sample, enabling improved measurements of spectral parameters, the detection of absorption features, and, most importantly, detailed timing studies. As shown in Figure 2, for a typical thermally emitting INS discovered by eROSITA (J1347) and a high-likelihood (but much fainter) XMM-Newton candidate (J1754), WFI observations with exposures of $\lesssim 200$ ks will enable the detection of low-amplitude, approximately sinusoidal pulsations. At fluxes $\gtrsim 10^{-13} \text{ erg cm}^{-2} \text{ s}^{-1}$ (0.2–2 keV), NewAthena will enable spin-period measurements for most eROSITA-discovered INSs, extending timing studies to fluxes approximately one order of magnitude fainter than those accessible with XMM-Newton.

The sub-arcsecond positional accuracy of WFI, combined

with the large collecting area of NewAthena, will be crucial for further XDINS searches. Most XDINS candidates are expected to remain unidentified in current eRASS and XMM-Newton catalogues, down to fluxes of $\sim 10^{-14}$ erg cm $^{-2}$ s $^{-1}$ and $(2.3-8)\times 10^{-16}$ erg cm $^{-2}$ s $^{-1}$, respectively. The limited number of detected counts leads to large positional uncertainties, making candidate identification particularly challenging in crowded fields such as the Galactic plane. Their nature is also difficult to confirm with current facilities, as long exposures are required for reliable localisation and characterisation (e.g. $\gtrsim 15-30$ ks with Chandra for accurate positions at $F_X \sim 10^{-13}$ erg cm $^{-2}$ s $^{-1}$). WFI will enable basic characterisation of these candidates with 1–2 ks exposures, allowing efficient confirmation or rejection of INS candidates.

4. Magnetars

Despite substantial observational and theoretical progress, several fundamental questions about magnetar physics remain open, on which NewAthena can provide important observational constraints. First, the mechanisms that trigger magnetar outbursts and govern the release and transport of magnetic energy remain uncertain. A central issue is the relative contribution of heat deposited within the crust and surface heating by particles returning along current-carrying magnetospheric field lines. Tracking the thermal and magnetospheric evolution throughout an outburst can constrain their relative contributions. Second, the strength and topology of the magnetic field near the stellar surface remain poorly constrained. Tracking the energies, widths and depths of magnetic-field-sensitive spectral features across rotational phase can probe the local field strength and constrain the geometry of the emitting region and surrounding magnetic structures (Tiengo et al., 2013; Kırımızbayrak and Heyl, 2023). Third, the role of non-dipolar and toroidal magnetic-field components, particularly those in the stellar interior, in governing long-term thermal evolution and activity remains poorly understood. Three-dimensional calculations show that Hall evolution can generate strong, localised, and non-axisymmetric magnetic structures, and that the surface-temperature distribution and expected activity rate depend strongly on the initial magnetic energy stored in non-dipolar components (Gourgoullos et al., 2016; De Grandis et al., 2022b; Dehman et al., 2023; Ascenzi et al., 2024). By constraining quiescent luminosities and surface-temperature anisotropies across a uniformly characterised sample, NewAthena can test competing magnetothermal evolutionary pathways and clarify the connections between magnetars, high-field pulsars, and thermally emitting INSs. Finally, it remains unclear whether Galactic magnetars are representative of the broader population, or whether the occurrence and energetics of outbursts, giant flares, and FRB-like radio activity vary with the environment. Follow-up of magnetar transients in nearby galaxies can link their X-ray properties to secure distances, stellar populations and host-galaxy conditions.

4.1. Outbursts: energy release and magnetic structure

4.1.1. Crustal and magnetospheric heating

Two principal heating channels may power the enhanced thermal emission during magnetar outbursts. In the crustal-heating scenario, magnetic dissipation or crustal failure deposits energy below the stellar surface (e.g. Li et al., 2016). The subsequent evolution depends on the depth and geometry of the energy deposition, anisotropic heat transport through the magnetised crust, and neutrino losses (e.g. De Grandis et al., 2025). In the magnetospheric-heating scenario, crustal motion twists a bundle of external field lines; charged particles carrying the associated currents return to the star and deposit their energy at the surface footpoints of the bundle (e.g. Beloborodov, 2009). The two channels can operate together: the same episode of crustal yielding may heat the crust while injecting a magnetospheric twist (Li et al., 2016; Coti Zelati et al., 2018). Their relative contributions must therefore be constrained by following the coupled evolution of the surface and magnetosphere from the outburst onset into the faint decay.

In crustal-heating models, shallow energy deposition controls primarily the peak and early decay, whereas deeper heating governs the longer-timescale evolution and can produce late-time shoulders or plateaus (e.g. De Grandis et al., 2025). The rise time and timing of the peak also constrain the duration of the heating episode, making prompt observations particularly valuable. In the calculations of De Grandis et al. (2025), reproducing observed peak luminosities requires at least part of the energy to be deposited at densities $\rho \lesssim 10^8$ g cm $^{-3}$, although this does not by itself distinguish shallow crustal heating from surface heating by return currents. Conversely, in simple untwisting-bundle models, the current-carrying footpoint contracts as the magnetospheric currents dissipate and, for approximately fixed twist amplitude and discharge voltage, its thermal luminosity is expected to scale roughly as the square of the hot spot area (Beloborodov, 2009). Compared with crustal heating models, the observational predictions of magnetospherically driven heating have been explored less systematically. Enlarging the outburst sample and identifying systematic trends could provide important empirical guidance for developing such models.

Multi-epoch, phase-resolved spectroscopy can test these predictions by tracking the temperature, emitting area and bolometric luminosity of individual surface components together with magnetospheric-scattering signatures and pulse morphology. The most direct discriminant may be the evolution of the inferred emitting area and its relation to luminosity. Crustal-heating models predict an evolution set by the lateral and radial diffusion of heat through the crust, whereas simple untwisting-bundle models predict a progressively contracting current-carrying footpoint, with a characteristic relation between its area and luminosity. Comparing the observed evolution of the inferred emitting area and luminosity with these model predictions, after accounting for absorption and the adopted spectral decomposition, can therefore constrain the relative contributions of crustal and return-current heating.

Existing facilities generally cannot resolve energy, rotational phase and outburst time simultaneously over the full dynamic

range of an outburst. NewAthena will measure these diagnostics repeatedly from the bright onset into the faint decay. Across a systematic sample, the geometry and evolution of the heated regions will also probe the hidden magnetic reservoir: models with similar spin-down dipoles can predict crustal-failure rates differing by orders of magnitude depending on the magnetic-energy distribution within the star (e.g., Bransgrove et al., 2018).

These observations can be confronted with three-dimensional magnetothermal frameworks such as MATINS and PARODY (De Grandis et al., 2022b; Dehman et al., 2023; Ascenzi et al., 2024). MATINS can convert surface-temperature maps into phase-resolved spectra and pulse profiles (Ascenzi et al., 2024), but end-to-end predictions for evolving outbursts remain to be developed. Equilibrium calculations have begun linking internal and external magnetic configurations, although their time-dependent coupled evolution remains an open challenge (Ntotsikas and Gourgouliatos, 2025). NewAthena will provide the phase-resolved constraints to discriminate among these models.

4.1.2. Phase-resolved magnetic-field tomography

Spin-down measurements constrain mainly the large-scale dipolar field and can thus miss much stronger, small-scale multipolar structures near the stellar surface. Mapping this local field is essential for understanding the spatial distribution of magnetic stresses, anisotropic heat transport and magnetospheric currents in the strongest magnetic fields known. Phase-resolved spectroscopy offers one of the few direct probes: if a feature is cyclotron in origin, its centroid traces the local field strength as the star rotates, while its width, depth and visibility constrain field gradients, line-forming geometry and strong-field radiative-transfer effects. The clearest example to date is the V-shaped spectral feature discovered with XMM-Newton during the outburst of SGR 0418+5729. Its rapidly changing centroid energy was interpreted as tracing a highly non-uniform local magnetic field close to the stellar surface (Tiengo et al., 2013). However, photon statistics still limit simultaneous constraints on line energy, width, and depth needed to robustly map the field-line geometry, determine whether comparable tracers can be recovered in faint outbursts, and extend local-field tomography beyond bright phases.

NewAthena will change this regime by combining unprecedented soft-X-ray throughput with high-count-rate timing. To demonstrate this, we used the 2009 XMM-Newton observation of SGR 0418+5729, obtained at an observed 0.3–10 keV flux of $\approx 6.5 \times 10^{-12} \text{ erg cm}^{-2} \text{ s}^{-1}$, as a template (Tiengo et al., 2013). Following Tiengo et al. (2013), we extracted 50 phase-resolved EPIC-pn spectra and fitted them in XSPEC with an absorbed continuum model, adding a cyclotron absorption component where required. From these fits, we took the recovered line centroid, width and depth over rotational phases $\phi = 0.10\text{--}0.30$, then reduced the continuum flux by a factor of ten, and simulated a 50-ks WFI observation in XSPEC.

Figure 3 shows that the normalised energy–phase map retains the absorption trough, visible as a photon deficit tracing the V-shaped feature between phases $\sim 0.1\text{--}0.3$. A representative continuum-only fit leaves structured residuals that are suppressed by adding a cyclotron component, while the phase

dependence of the line centroid, width and depth is recovered. WFI can therefore extend local-field tomography to outbursts an order of magnitude fainter than the event in which the feature was discovered, enabling tighter constraints on the underlying magnetic-field geometry.

4.2. Quiescent thermal anisotropy

Whereas transient activity can probe rapid rearrangements of the local magnetic field, quiescent emission traces the long-term thermal imprint of its evolution. Surface-temperature anisotropies retain information about heat transport and the internal field, but current observations generally lack the sensitivity for phase-resolved spectroscopy in quiescence (e.g. Ascenzi et al., 2024). Moreover, an unresolved complex temperature structure can bias the temperatures and emitting areas inferred from phase-averaged, single-component models. Resolving this structure is thus essential for testing magneto-thermal evolution.

To assess whether WFI can resolve such thermal variations, we simulated emission from a cool stellar surface plus a localised hot region at faint flux levels, modelling both components as blackbodies. The cool component had temperature $kT_{\text{cool}}^{\infty} = 0.1 \text{ keV}$ and apparent radius $R_{\text{cool}}^{\infty} = 12 \text{ km}$, while the hot region had reference values $kT_{\text{hot}}^{\infty} = 0.3 \text{ keV}$ and $R_{\text{hot}}^{\infty} = 0.4 \text{ km}$. We adopted a distance of 5 kpc, a hydrogen absorption column density of $N_{\text{H}} = 5 \times 10^{21} \text{ cm}^{-2}$, and a phase-averaged observed 0.3–10-keV flux of $5 \times 10^{-14} \text{ erg cm}^{-2} \text{ s}^{-1}$. We assumed a spin period of 10 s and a sinusoidal pulse profile with a fractional semi-amplitude of 50%. Rather than modelling a specific viewing geometry, we prescribed the phase dependence phenomenologically. We imposed a sinusoidal variation in the temperature of the hot component of $\pm 10\%$ around its reference value. At each rotational phase, we then adjusted the normalisation of the hot component so that the combined emission from the cool and hot components reproduced the assumed pulse profile. The cool surface remained constant. We generated 50 independent realizations of a 100-ks WFI-LDA observation, each folded into eight 12.5-ks phase bins. Spectra were simulated within XSPEC and jointly fitted over 0.3–3 keV with the C-statistic. The cool-component parameters were tied across phase, whereas the hot-component temperature and normalisation were allowed to vary in each bin. This choice was made to test whether WFI could recover the imposed phase-dependent spectral variations without assuming their sinusoidal form in the fit.

Figure 4 compares the injected phase dependence with the median recovered values and central 68% intervals across the 50 realizations. Typical fractional statistical precisions on the recovered quantities are $\approx 3\%$ for the hot-spot temperature, $\approx 7\%$ for the apparent radius and $\approx 3\%$ for the hot-component flux. The steady surface component is recovered at $kT_{\text{cool}}^{\infty} = 100.3^{+0.4}_{-1.1} \text{ eV}$ and $R_{\text{cool}}^{\infty} = 12.1^{+0.4}_{-0.3} \text{ km}$. As a sensitivity check, we refitted the same 50 realizations allowing a single N_{H} , tied across all phase bins, to vary freely while retaining the adopted distance and emission model. The median recovered values and central 68% intervals then become $kT_{\text{cool}}^{\infty} = 100.4^{+2.2}_{-2.9} \text{ eV}$ and $R_{\text{cool}}^{\infty} = 12.0^{+1.9}_{-1.5} \text{ km}$. We remark that

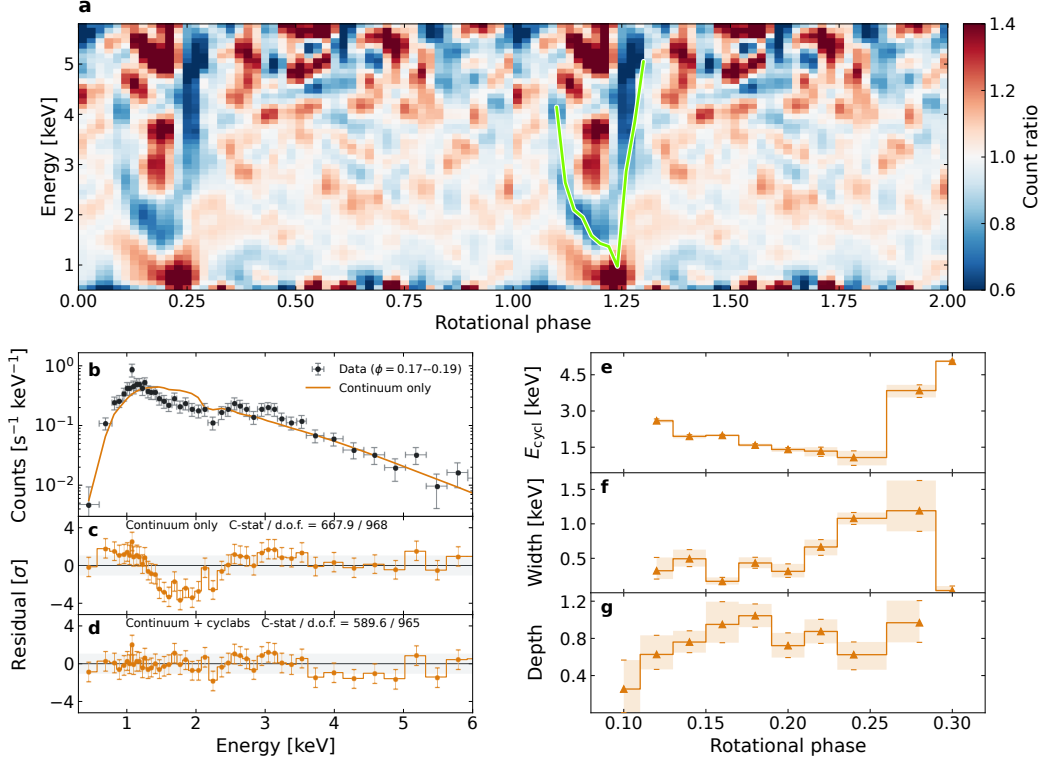


Figure 3: Phase-resolved cyclotron-line tomography of a magnetar observed at a 0.3–10 keV flux of $6.5 \times 10^{-13} \text{ erg cm}^{-2} \text{ s}^{-1}$, with absorption-feature properties matching those measured from SGR 0418+5729 during its outburst. Panel a: phase- and pulse-normalised count-ratio map from a 50-ks WFI simulation. The green solid curve in the second cycle traces the injected line-centroid evolution. Panel b: representative phase-resolved spectrum. Panels c,d: residuals from the continuum-only and continuum-plus-cyclabs fits, respectively; grey bands indicate $\pm 1\sigma$. Panels e–g: best-fitting line centroid, width and depth, with nominal 1σ uncertainties ($\Delta C = 1$ for one parameter of interest) shown by error bars and orange shading. Some error bars are smaller than the markers. The centroid and width at $\phi = 0.10$ are omitted because valid two-sided uncertainty bounds could not be obtained. The depth at $\phi = 0.30$ is omitted because the fit reaches its imposed upper bound.

real NS surfaces are likely to exhibit more complex, continuous temperature distributions than the one adopted here. Nevertheless, the substantially improved phase-resolved constraints enabled by WFI will provide a much more sensitive probe of the underlying thermal structure. Applied across sources, these measurements will test how the hidden internal field governs long-term heat transport and thermal evolution.

4.3. Extragalactic transients and radio activity

Galactic magnetars remain the main laboratories for detailed crust–magnetosphere physics, but nearby galaxies provide access to a much larger population at known distances and in characterisable host environments. Detecting magnetar transient activity beyond the Milky Way would constrain their luminosity function, recurrence rate and active lifetime, and test dependencies on star formation, stellar age, metallicity and environment.

We assessed the detectability and characterisation of extragalactic magnetar outbursts using a two-blackbody spectrum with $kT_{\text{cool}} = 0.35 \text{ keV}$ and $kT_{\text{hot}} = 0.80 \text{ keV}$, contributing 35% and 65% of the intrinsic 0.3–10 keV flux, respectively (Coti Zelati et al., 2018). We varied the total flux over absorbing columns spanning $N_{\text{H}} = 10^{21}–10^{22} \text{ cm}^{-2}$. Figure 5 distinguishes the requirements for source detection, spectral characterisation and blind pulsation searches. In a 100-ks WFI observation, the unabsorbed 0.3–10 keV detection threshold is $\approx (3–7) \times 10^{-16} \text{ erg cm}^{-2} \text{ s}^{-1}$, whereas distinguishing the two-temperature spectrum from a single-temperature model requires $\approx (2–5) \times 10^{-15} \text{ erg cm}^{-2} \text{ s}^{-1}$, comparable to that needed to constrain the hot component temperature and normalisation. At 3 Mpc, these correspond to $\approx (3–8) \times 10^{35} \text{ erg s}^{-1}$ for detection and a few $10^{36} \text{ erg s}^{-1}$ for spectral characterisation. Blind

searches for 30% fractional-rms pulsations at periods of 2–20 s require $\approx (3–7) \times 10^{-15} \text{ erg cm}^{-2} \text{ s}^{-1}$, favouring Local Group sources unless the outburst is exceptionally luminous.

Beyond outbursts, rapid NewAthena follow-up of well-localised giant-flare candidates in nearby galaxies could search for enhanced X-ray emission following the initial event. Candidate samples suggest only a few such opportunities per decade within $\approx 5 \text{ Mpc}$, although the rate remains uncertain (Burns et al., 2021). At these distances, NewAthena observations would be sensitive primarily to luminous post-flare emission. A detection would allow us to investigate the spectral and temporal evolution of the X-ray counterpart, whereas non-detections would constrain the presence of sustained luminous emission following the flare.

Coordinated X-ray and radio observations of a nearby, localised repeating FRB could determine whether persistent or burst X-ray emission is present down to the WFI sensitivity limit and, for sufficiently close sources, characterise its spectrum and temporal behaviour. Simultaneous coverage could then test whether coherent radio bursts coincide with enhanced surface heating or magnetospheric reconfiguration. SGR J1935+2154 established that FRB-like radio bursts can accompany enhanced magnetar activity (e.g., CHIME/FRB Collaboration, 2020). By measuring persistent luminosity, spectrum and pulse morphology during radio-active intervals, NewAthena can test whether the persistent X-ray emission brightens around the time of a radio burst, even when no prompt X-ray burst is detected.

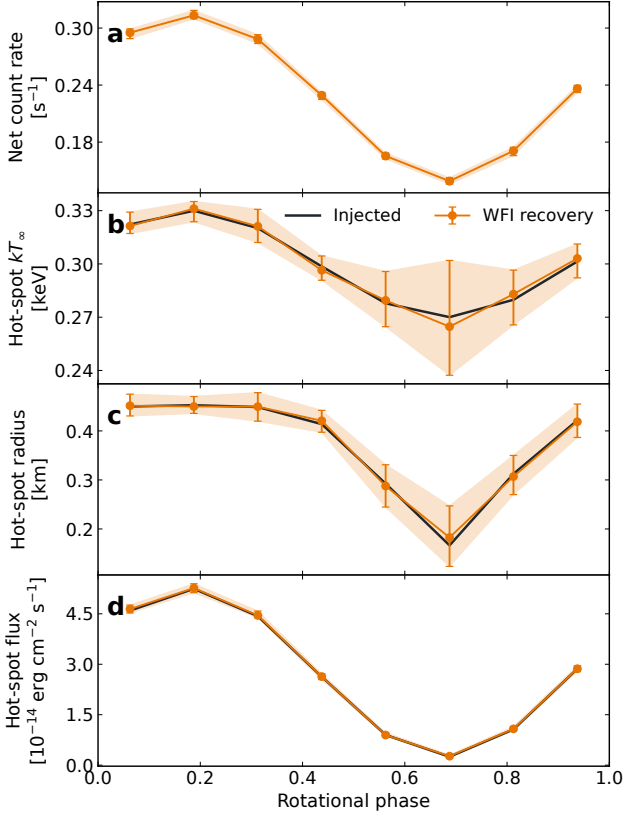


Figure 4: Simulated phase-resolved spectral variability of a 100-ks WFI observation assuming a cool surface, a localised hot spot and a phase-averaged flux of $5 \times 10^{-14} \text{ erg cm}^{-2} \text{ s}^{-1}$ (0.3–10 keV). Panel **a**: recovered pulse profile; panels **b–d**: injected and recovered hot-component temperature, apparent radius and observed flux for an injected 10% temperature semi-amplitude. Orange points and bands show the median and central 68% intervals from 50 complete Monte Carlo realizations; black curves show the injected phase dependence. Absorption was fixed, while the cool-component temperature and normalisation were fitted jointly and tied across phase.

5. Central compact objects

The combination of weak spin-down-inferred dipole magnetic fields and relatively bright, localised thermal emission makes CCOs an especially puzzling population of INSS. They occupy a sparsely populated region of the $P - \dot{P}$ diagram (see Fig. 1), raising questions about their magnetic field configuration and evolution. Glitches or comparably large timing noise, reported for two CCOs (e.g. Perez et al., 2026), are also consistent with the young age of these INSS. For PSR J1210–5226 (1E 1207.4–5209), an electron cyclotron resonance interpretation of several X-ray absorption lines implies a dipole field strength comparable to that inferred from the spin-down parameters. For the same object, the recent study by Zhang et al. (2026) reported on the detection of faint, stable radio pulsations, while also establishing deep upper limits on radio emission from three other CCOs. Although the morphology and polarization of the radio pulse from PSR J1210–5226 are broadly consistent with those of ordinary pulsars, its X-ray behaviour is not. In addition to the absence of detectable non-thermal X-ray emission, CCOs exhibit very small hot spot areas.

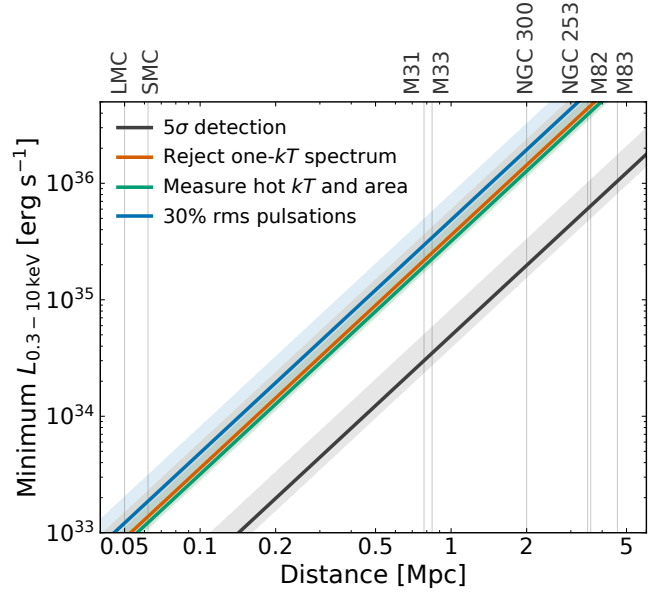


Figure 5: WFI 100-ks reach for ordinary magnetar outbursts in nearby galaxies. The curves give the minimum 0.3–10 keV luminosity as a function of distance for a 5σ source detection, targeted rejection of a one-temperature spectrum, measurement of the hot component’s temperature and area, and detection of 30% fractional-rms pulsations in a blind 2–20-s search. The same-colour bands span $N_H = 10^{21} - 10^{22} \text{ cm}^{-2}$. Vertical lines mark representative nearby galaxies.

Several models have been proposed to explain the observed properties of the CCOs. One model assumes intrinsically weak dipole fields. A highly tangled internal magnetic field, with a weak or absent dipole component, could produce small-scale strong multipolar components at the surface, producing the small hot spots (Igoshev et al., 2021) and possibly the radio emission (Zhang et al., 2026). Another model proposes early burial of (relatively strong dipolar) magnetic fields under supernova fallback material and reemergence of these magnetic fields over time (Viganò and Pons, 2012; Ho, 2015). Such magnetic field evolution has also been proposed to reconcile the very large and unphysical characteristic ages inferred for CCOs from their spin parameters with the much younger ages of their host SNRs (Rogers and Safi-Harb, 2016, 2017). On-going low-level accretion from fallback material has also been discussed, potentially explaining the X-ray hot spots, X-ray timing variability, and low radio luminosities (e.g. Perez et al., 2026). Any successful model must address the challenging question of what and where the CCO descendants are. Considering their abundance in SNRs, CCOs may constitute a significant fraction of the NS population, yet searches for their immediate descendants have been so far unsuccessful (Gotthelf et al., 2013). After the associated SNR has faded, the thermal X-ray emission from the cooling NS should persist, while faint radio emission together with its spin-down properties could lead to its misidentification as a very old pulsar that is anomalously too hot in X-rays for its inferred age. A previously promising CCO-descendant candidate, the hot X-ray pulsar Calvera, has now been identified as an actual CCO associated with a faint SNR, with an inferred dipole magnetic field larger by a factor of $\sim 5 - 15$ than those of the

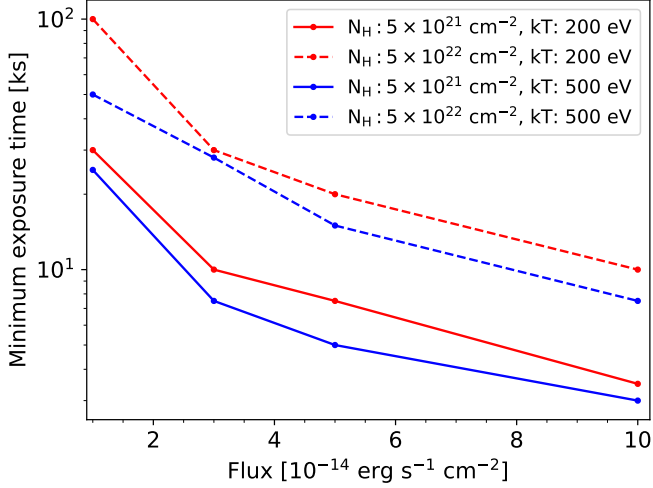


Figure 6: Minimum NewAthena WFI exposure required to distinguish the simulated CCO-like blackbody spectra from the adopted steep power-law model, as a function of observed 0.2–12 keV flux. The blackbody models have $N_{\text{H}} = 5 \times 10^{21}$ or $5 \times 10^{22} \text{ cm}^{-2}$ and $kT_{\text{BB}} = 200$ or 500 eV . The power-law model has $N_{\text{H}} = 5 \times 10^{22} \text{ cm}^{-2}$ and $\Gamma = 3$. See text for the model-selection criterion.

other known CCOs (Rigoselli et al., 2024).

The CCOs’ unperturbed thermal X-ray emission provides a rare opportunity to search for a notable evolution of their surface temperature distribution. CCOs are young, and NS cooling – which is dependent on the interior composition and physical conditions – is expected to proceed rapidly at early ages. Thus, very young CCOs are ideal targets to monitor NS cooling on human timescales and probe the physics of the NS interior. The youngest CCO currently known, the Cas A CCO, enabled measurements of a decline in the thermal X-ray flux over about two decades (Posselt and Pavlov, 2022; Zhao et al., 2025). The surface or atmosphere composition of CCOs is still debated (e.g. Alford and Halpern, 2023) and can be constrained with sensitive X-ray spectra. The long-term thermal X-ray monitoring of the same object can then probe the microphysics of the NS core, including the onset of neutron superfluidity and associated enhancement of neutrino emission through Cooper-pair formation (e.g. Shternin et al. 2011).

NewAthena’s high soft X-ray sensitivity, timing capability, and WFI’s large grasp will be instrumental in addressing the CCO puzzle on various fronts:

1. **Identify new CCOs and their descendants:** NewAthena’s WFI will enable systematic searches for thermal X-ray sources in SNRs, or probe faint radio pulsars deep enough for their X-ray emission in order to identify CCOs and CCO descendants. Gotthelf et al. (2013) estimated that there should be ~ 100 CCO descendants in the volume sampled by the radio surveys at that time. This volume and sensitivity to fainter pulsars are expected to have strongly increased at the time of NewAthena’s operation due to already ongoing and planned future pulsar surveys. The all-sky SKA pulsar survey, for instance, is expected to at least triple the currently known number of non-recycled pulsars (Keane et al., 2026). At the same time, many new SNRs

are identified due to new radio imaging surveys (Anderson et al., 2025). NewAthena will be able to quickly “scan” these new SNRs as well as efficiently probe suitable radio pulsars to solve the puzzle about the role of CCOs within the overall NS population. To illustrate NewAthena’s detection and identification potential, we simulated multiple simple CCO spectra with the WFI instrument response. Blackbody fit temperatures of $kT_{\text{BB}} \sim 200 - 700 \text{ eV}$ are reported for the currently known CCOs. The faintest of the CCOs has an X-ray flux of $\sim 10^{-13} \text{ erg cm}^{-2} \text{ s}^{-1}$. For our NewAthena WFI simulations, we adopted X-ray fluxes down to $\sim 10^{-14} \text{ erg cm}^{-2} \text{ s}^{-1}$, a factor of ten below that of the faintest known CCO. We used XSPEC to generate absorbed blackbody spectra ($N_{\text{H}} = (0.5-5) \times 10^{22} \text{ cm}^{-2}$ and $kT_{\text{BB}} = 200-500 \text{ eV}$) and absorbed power-law spectra ($N_{\text{H}} = 5 \times 10^{22} \text{ cm}^{-2}$ and $\Gamma = 3$). The steep power law was chosen as it may be the most difficult to distinguish from the strongly absorbed blackbody-like spectra of CCOs. We then applied Bayesian X-ray Analysis to compare the goodness of blackbody and power law fits to the simulated spectra via the resulting Bayes factors. For this, we estimated the minimum exposure time at which the chance to obtain a Bayes factor indicating the wrong source nature is less than 5%. The results are presented in Fig. 6 and illustrate that less than 30 ks (100 ks) are sufficient to confidently establish the source nature of medium (strongly) absorbed CCOs in the probed flux range.

2. **Pulsation search, phase-resolved spectra, and spin-down measurements:** Only three of the known CCOs have measured pulsations and spin-down rates. Their location near SNR centres introduces non-trivial or non-negligible X-ray background, making high sensitivity and adequate spatial resolution essential for detecting moderate or low pulsed fractions. NewAthena will probe lower pulsed fractions in known CCOs, search for pulsations in newly identified CCOs and CCO-descendant candidates, and measure period derivatives in order to constrain their spin-down power and infer surface dipole magnetic fields. Phase-resolved spectroscopy will further constrain the surface temperature distribution and provide information on the surface composition.
3. **Monitoring for timing and spectral variability:** The surprising glitch of PSR J1210–5226 and the reported flux evolution of the Cas A CCO highlight the need for CCO monitoring campaigns. Thanks to NewAthena’s high sensitivity, such studies can be carried out in reasonable exposure times. Accurate and stable spectral calibration will be essential for connecting NewAthena’s measurements to earlier observations and minimising uncertainties associated with cross-calibration between instruments.
4. **Constrain the CCO age and progenitor from their SNR association:** SNRs can be studied in great detail with NewAthena thanks to the large field of view of WFI and the spectral resolution of X-IFU. Measurements of the remnant dynamics, plasma properties, and ejecta abundances

can constrain both the ages and explosion properties of CCO-hosting SNRs, while comparison of the ejecta composition with nucleosynthesis models can probe the masses and nature of their supernova progenitors (e.g. Braun et al., 2023).

6. Summary

NewAthena will place RPPs, XDINSs, magnetars, and CCOs within a single observational framework of comparable sensitivity and precision. Its combination of large effective area, high spectral resolution, and low background will push observations of XDINSs and CCOs to much fainter fluxes, and will provide, for the first time, timing and spectral measurements of equivalent quality across all the different classes of INSs. This will make it possible to test magneto-thermal evolutionary models across the population as a whole, rather than source by source (Viganò et al., 2013).

High-resolution phase-resolved spectroscopy will be central to this progress. In magnetars, it will constrain the local magnetic field through cyclotron features and trace the geometry and evolution of hotspots during outbursts, disentangling the relative roles of crustal and magnetospheric heating (Sect. 4.1–4.2). In RPPs, the same approach will test whether small, strongly magnetised polar caps are a generic feature of pulsar magnetospheres, with direct consequences for pair-cascade and radio-emission models (Sect. 2). A high-quality phase-resolved X-ray spectra from NewAthena will be highly complementary to existing Fermi LAT gamma-ray and radio data helping to complete a unified picture of magnetosphere and its interaction with the NS surface. In CCOs, deeper pulsation searches and improved spin-down measurements will help establish their place within the NS population and the reach of their link to young magnetars (Sect. 5).

Moreover, the strong constraints on stellar properties and thermal evolution provided by NewAthena observations of INSs will be compared with theoretical cooling models to search for signatures of dark matter and physics beyond the Standard Model. In fact, compact stars provide a unique laboratory for probing dark matter through their thermal evolution and observable properties. Dark matter particles can either accumulate over the stellar lifetime and/or be produced within the star, affecting the matter distribution, mass–radius relation, and cooling (e.g. Sagun et al., 2023). Heavy dark matter accumulated in the NS core deepens the gravitational potential, triggering fast neutrino-emission processes at lower stellar masses and thereby accelerating cooling (Ávila et al., 2024). Axions and other weakly interacting light particles produced within the star can provide an additional cooling channel if they escape freely (e.g., Buschmann et al. 2022).

Given the high sensitivity and large field-of-view of WFI, serendipitous discoveries of INSs, in, e.g., wide-area galactic surveys are also possible. One promising type is older, mostly-dormant and very faint but occasionally flaring magnetars some of which may be surrounded by more persistent nebulae (e.g., Swift J1834.9-0846; Kargaltsev et al. 2012b). It may also be possible to finally discover older NSs (e.g., slowly rotating RPPs

with shut-off pair cascades) via emission caused by weak accretion from ISM (see Afonina et al. 2026 and references therein). Thus, surveys of the Galactic plane, also hosting a variety of other interesting high-energy sources, should be encouraged.

Overall, by combining unprecedented sensitivity, spectral resolution, and timing capabilities, NewAthena will help to reveal how the diversity of X-ray properties observed across the INS population encodes their birth conditions through their subsequent magneto-thermal evolution.

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